

Algorithms 2

Graph Colouring

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1. Colourings in Graphs

Definition 1.1.1 An colouring of a graph G with k colours is a function $c : V(G) \rightarrow [k]$, satisfying that $\forall uv \in E(G)$ it holds that $c(u) \neq c(v)$.

Definition 1.1.2 A graph G is said to be *k -colourable* if there is a graph colouring on G using at most k distinct colours. i.e. $\forall v : c(v) \in [k]$.

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Definition 1.1.3 Let $\chi(G)$ be the minimum $k \in \mathbb{N}$ for which G is *k -colourable*.

Remark

Verifying that G is *k -colourable* for $k \geq 3$ is **NPC**.

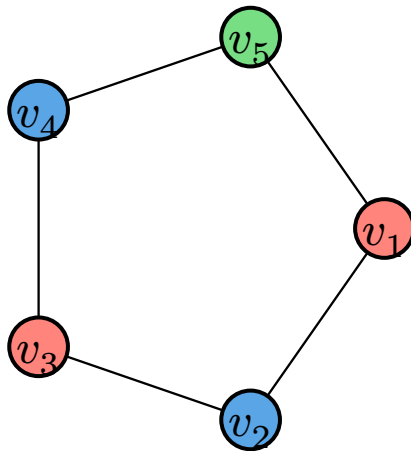
2. A Greedy Colouring Algorithm

- A Greedy Colouring Algorithm:
 - Input:
 - A graph G
 - An ordering $v_1, v_2, \dots, v_n$ of $V(G)$
 - Output:
 - A legal vertex-colouring of G
1. Scan the vertices according to the ordering.
 2. When considering v_i , assign it the least available colour.
 3. This always produces a legal colouring.

Question 2.1.1 How useful is this algorithm for bounding $\chi(G)$?

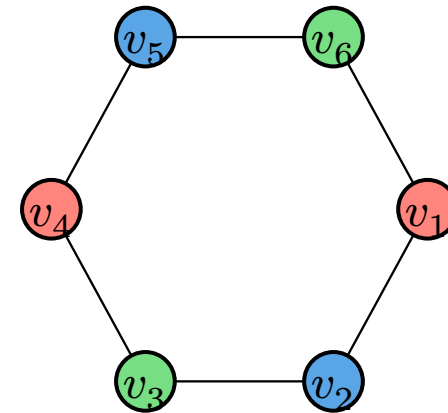
Question 2.2.1 How useful is this algorithm for bounding $\chi(G)$?

C_5 (greedy uses 3 colours)



v_1, v_2, v_3, v_4, v_5
1, 2, 1, 2, 3

C_6 (greedy can also use 3 colours)



$v_1, v_4, v_2, v_3, v_5, v_6$
1, 1, 2, 3, 2, 3

- For every graph G , define

$$\Delta(G) := \max\{\deg_{G(v)} : v \in V(G)\}$$

Theorem 2.3.1 $\chi(G) \leq \Delta(G) + 1$

Proof: Every vertex sees at most $\Delta(G)$ distinct colours, so at least one colour is still *free*, and it will be chosen by the greedy algorithm.

- When the bound is tight?
 - There are graphs with

$$\chi(G) = \Delta(G) + 1$$

- Examples:
 - Odd cycles: $\Delta = 2$ and $\chi = 3$
 - Complete graphs: $\Delta(K_r) = r - 1$ and $\chi(K_r) = r$
- What about the rest? Can we do better?

3. Brooks' Theorem

Theorem 3.1.1 (Brooks' Theorem) Let G be a connected graph. Then

$$\chi(G) \leq \Delta(G)$$

unless G is complete or an odd cycle.

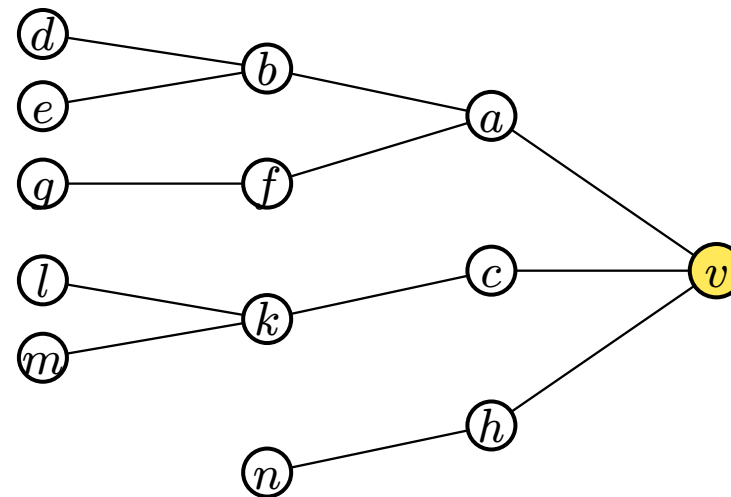
Observation 3.1.2 Brooks' upper bound can be far from optimal in general.

proof:

- Assume $\Delta(G) \geq 3$.
- If $\Delta(G) \leq 2$, then connected G is a path or an even cycle, so the theorem is immediate.
- Distinguish two cases:
 - There exists $v \in V(G)$ with $\deg(v) < \Delta(G)$.
 - Otherwise G is $\Delta(G)$ -regular.

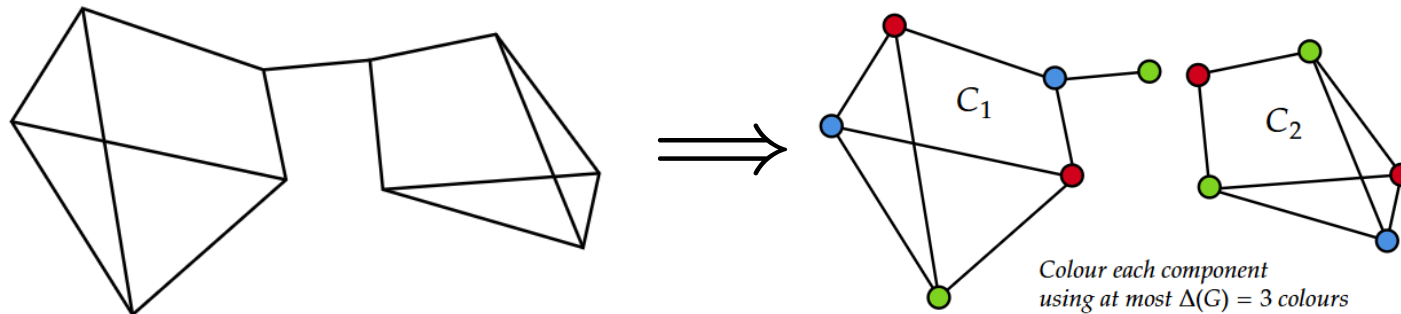
(We either choose an edge or we don't)

- Choose $v \in V(G)$ with $\deg(v) < \Delta(G)$.
- Let T be a spanning tree of G rooted at v .
- Order vertices so descendants appear before ancestors and v is last.
- In this ordering, each vertex except possibly v has at most $\Delta(G) - 1$ earlier neighbours.
- Each vertex has at least 1 available colour, v has one because it has at most $\Delta(v) - 1$ neighbours.
- We are left with a greedy colouring with at most $\Delta(G)$ colours.



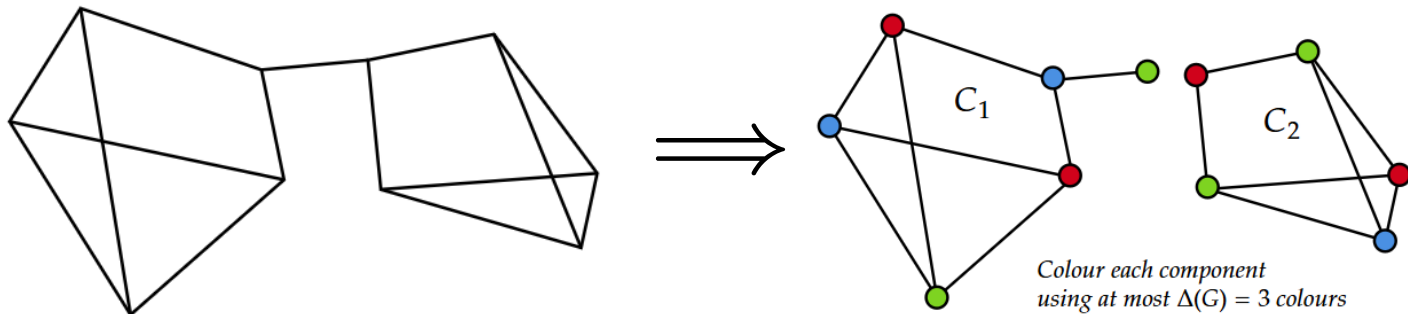
Example descendants-first order: $d, e, b, g, f, a, l, m, k, c, n, h, v$.

- Assume G is $\Delta(G)$ -regular.
- If $\kappa(G) = 1$, then G has a cut-vertex v .
- Break G into 2 components with v in both of them.
- In each component $\deg(v) < \Delta(G)$, colour each component using $\Delta(G)$ colours using Case 1.
3-regular graph, with vertex cuts.

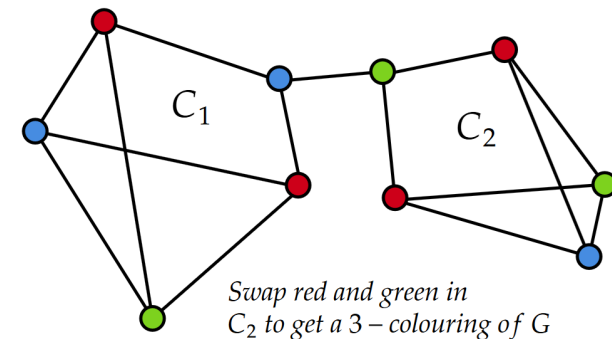


- It remains to join the parts. How?

- Assume G is $\Delta(G)$ -regular.
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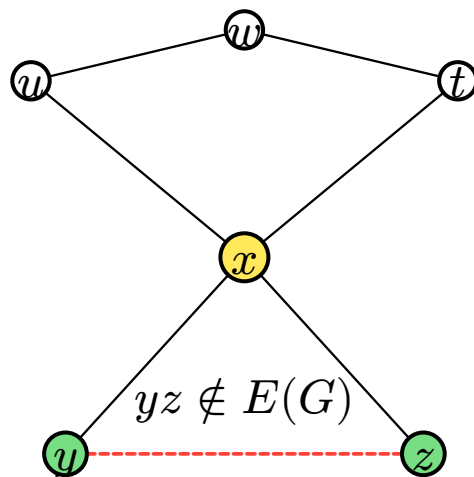


- It remains to join the parts. How?
- If $c(v) = i$ in C_1 , and $c(v) = j$ in C_2 . Swap in C_2 the colours of all vertices colour in j to i and all colours i to j .



- **Goal:** find $x, y, z \in V(G)$ such that:
 - $xy, xz \in E(G)$
 - $yz \notin E(G)$
 - $G - \{y, z\}$ is connected

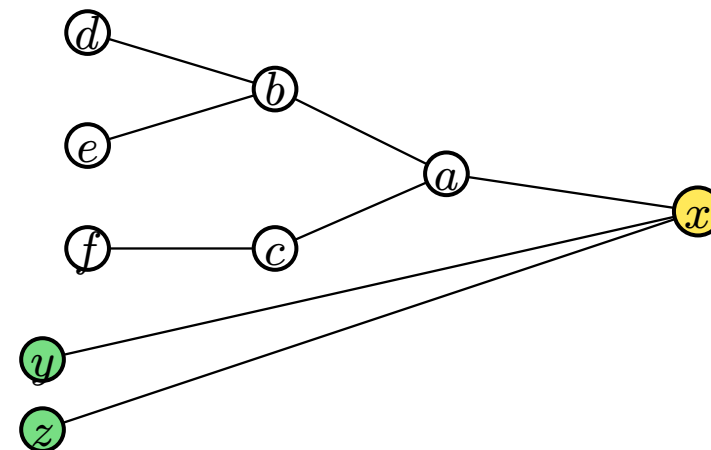
Cherry condition



$xy, xz \in E(G)$ and $G - \{y, z\}$ stays connected.

- If such vertices exist:
- Build a spanning tree of $G - \{y, z\}$ rooted at x .
- Use an order with x last and y, z first.
- Since y and z are nonadjacent, they may receive the same colour.
- Greedy colouring then gives a legal $\Delta(G)$ -colouring.

Rooted tree for greedy order



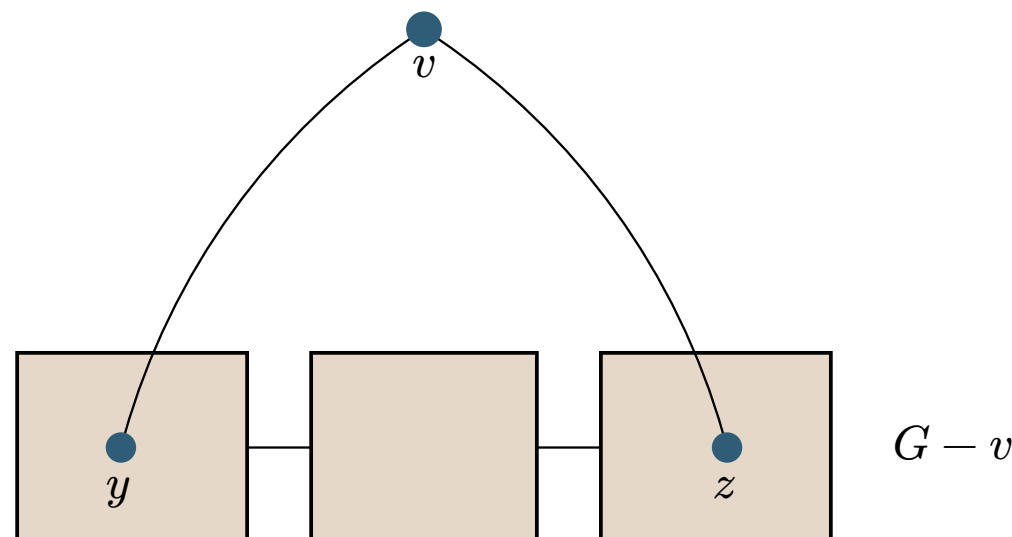
Order example: $y, z, d, e, b, f, c, a, x$ (with x last).

Lemma 3.4.1 (Structural Lemma) Let $3 \leq k \in \mathbb{N}$ and let G be a k -regular, 2-connected, non-complete graph. Then there exist $x, y, z \in V(G)$ such that:

1. $xy, xz \in E(G)$
2. $yz \notin E(G)$
3. $G - \{y, z\}$ is connected.

- **Subcase 1:** $\kappa(G) = 2 \implies$ there exists v such that $G - v$ is connected but not 2-connected.
 - Analyze the block tree of $G - v$.
 - Pick y, z in different leaf blocks, and set $x := v$.
 - Then $G - \{y, z\}$ stays connected.

(pick y, z that are not cut-vertices)



Block decomposition of $G - v$: blocks are separated; y, z are non cut-vertices in leaf blocks; y, z are connected to v .

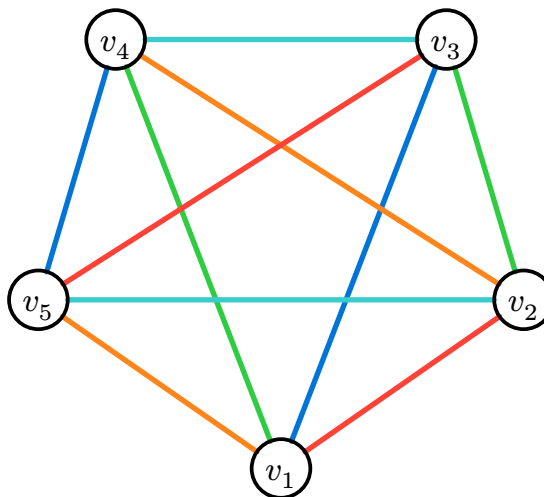
- **Subcase 2:** By assumption $\kappa(G - v) \geq \kappa(G) - 1 \geq 2$ for every $v \in V(G)$.
 - Since G is not complete, pick nonadjacent y, z among neighbours of some x .
 - $\kappa(G - y) \geq 2$
 - Connectivity conditions force $G - \{y, z\}$ to remain connected.

4. Edge Colouring

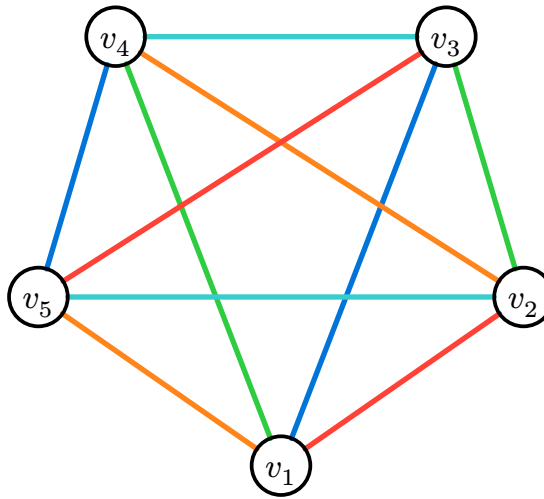
Definition 4.1.1 An edge-colouring of a graph G with k colours is a function

$$\varphi : E(G) \rightarrow [k]$$

Definition 4.1.2 An edge-colouring is called proper if incident edges receive distinct colours.



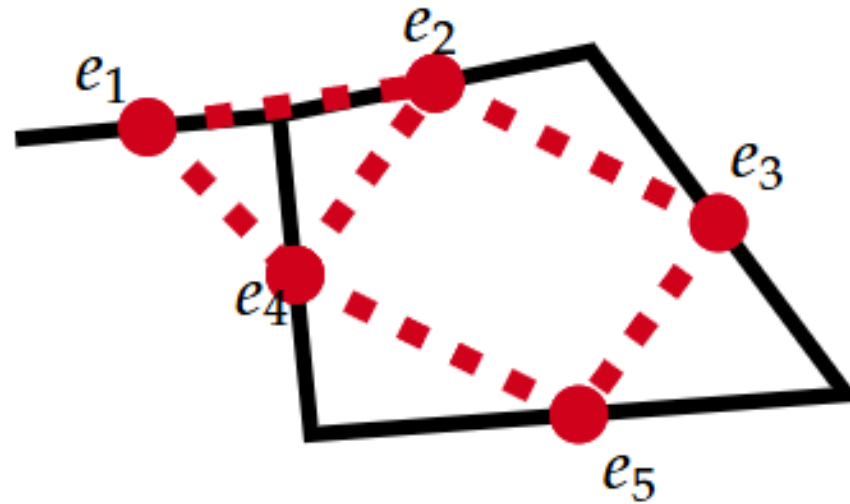
Definition 4.1.3 The chromatic index of G , denoted by $\chi'(G)$. Which is the least k such that G has a proper edge-colouring with k colours.



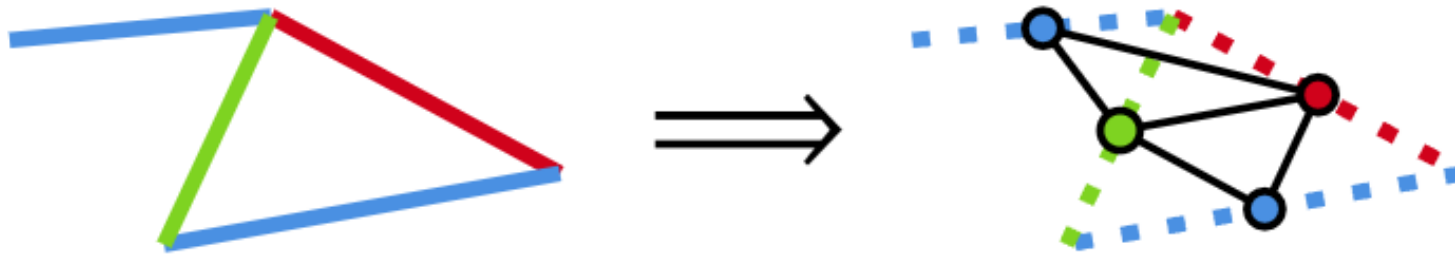
- here $\chi'(G) = 5$.

Definition 4.2.1 The line graph of G , denoted $L(G)$, is the graph with:

- $V(L(G)) = E(G)$
- two vertices adjacent iff the corresponding edges of G are incident.

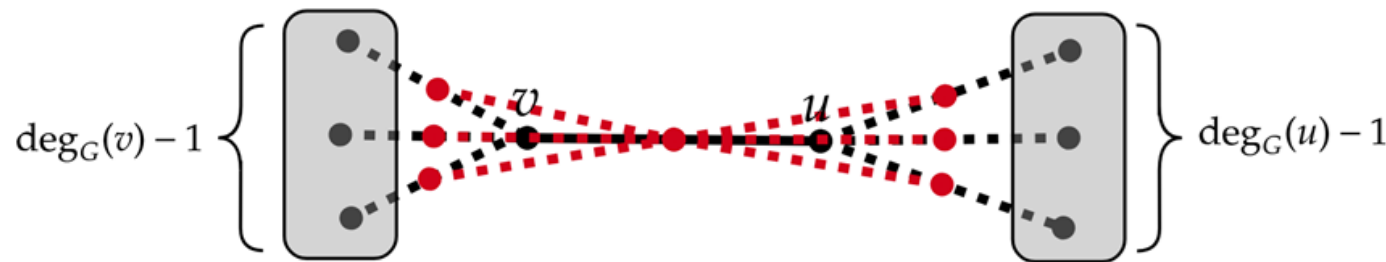


Observation 4.3.1 $\chi'(G) = \chi(L(G))$



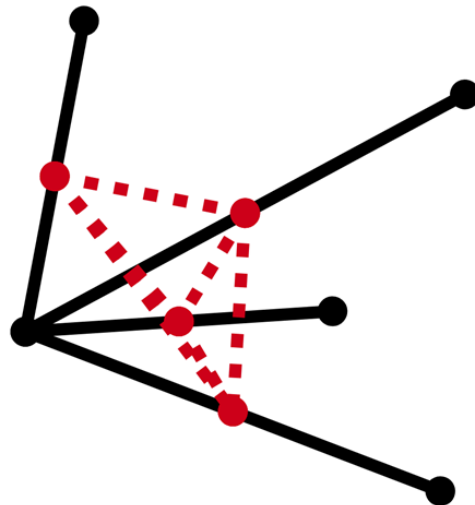
Observation 4.4.1 $\Delta(L(G)) \leq 2(\Delta(G) - 1)$

- Fix $e = uv \in E(G)$.
- $v_e \in V(L(G))$ is adjacent to all vertices corresponding to edges incident to u or v .
- So $\deg_{L(G)}(v_e) = \underbrace{(\deg_{G(u)} - 1) + (\deg_{G(v)} - 1)}_{\substack{\text{We do not count } e \text{ as an incident edge} \\ \text{to itself.}}} \leq 2(\Delta(G) - 1)$.
- We conclude that $\Delta(L(G)) \leq 2(\Delta(G) - 1)$.



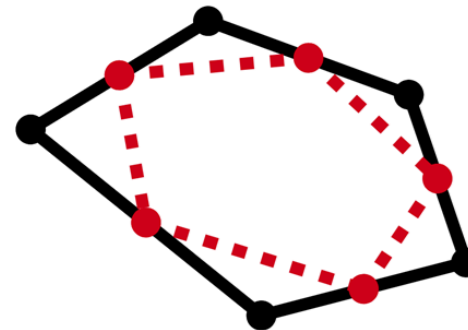
Observation 4.5.1 If $L(G)$ is a clique, then G is a star and $\chi'(G) = \Delta(G)$.

Observation 4.5.2 If $L(G)$ is an odd cycle, then G is also an odd cycle with $\chi'(G) = \Delta(G) + 1$.



$$L(K_{1,4}) \implies K_4$$

$$L(K_{1,k}) \implies K_k$$



Observation 4.5.3 $\Delta(G) \leq \chi'(G) \leq 2(\Delta(G) - 1)$

- $\Delta(G) \leq \chi'(G)$ for all G .
- If $L(G)$ is clique or an odd cycle, then $\chi'(G) \leq \Delta(G) + 1$.

Otherwise:

- $\chi'(G) = \chi(L(G))$
- $\Delta(L(G)) \leq 2(\Delta(G) - 1)$
- So that by Brooks:

$$\chi'(G) = \underbrace{\chi(L(G)) \leq \Delta(L(G))}_{\substack{\chi(G') \leq \Delta(G') \\ G' \text{ is not a clique or odd cycle}}} \leq 2(\Delta(G) - 1)$$
- We then get:

$$\Delta(G) \leq \chi'(G) \leq 2\Delta(G) - 2$$

Observation 4.5.3 $\Delta(G) \leq \chi'(G) \leq 2(\Delta(G) - 1)$

- $\Delta(G) \leq \chi'(G)$ for all G .
- If $L(G)$ is clique or an odd cycle, then $\chi'(G) \leq \Delta(G) + 1$.

Otherwise:

- $\chi'(G) = \chi(L(G))$
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- We then get:

$$\Delta(G) \leq \chi'(G) \leq 2\Delta(G) - 2$$

Can we do better?

5. Konig's Theorem

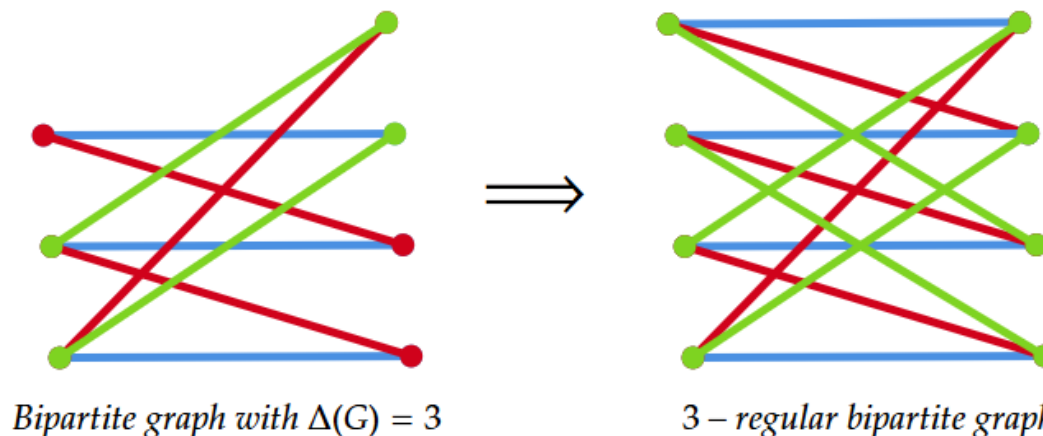
Theorem 5.1.1 (König's Theorem) If G is bipartite, then

$$\chi'(G) = \Delta(G)$$

- Remember that $\chi'(G) \geq \Delta(G)$ always holds, so we only need to show $\chi'(G) \leq \Delta(G)$ for bipartite graphs.

proof:

- Degree-regular bipartite graphs admit a perfect matching (See in the practice session).
- Removing edges of a perfect matching from a k -regular bipartite graph gives a $(k - 1)$ -regular bipartite graph.
- By induction on k , every k -regular bipartite graph is k -edge-colourable.
- Any bipartite graph can be embedded into a $\Delta(G)$ -regular bipartite supergraph.
- Restrict the colouring from the supergraph back to G .



6. Vizing's Theorem

Theorem 6.1.1 (Vizing's Theorem) For every simple graph G ,

$$\chi'(G) \leq \Delta(G) + 1$$

- Combined with the trivial lower bound:

$$\Delta(G) \leq \chi'(G) \leq \Delta(G) + 1$$

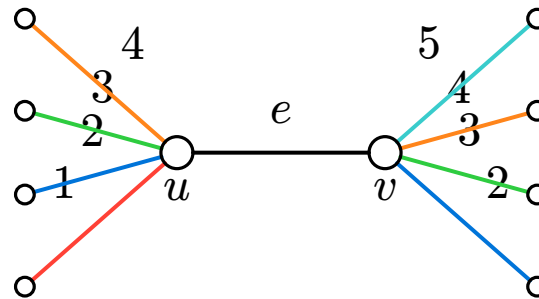
- So every graph is either:
 - Class 1 ($\chi'(G) = \Delta(G)$)
 - Class 2 ($\chi'(G) = \Delta(G) + 1$).
 - It is NP-hard to determine the class of a given graph.

Theorem 6.2.1 (Vizing's Theorem) For every simple graph G ,

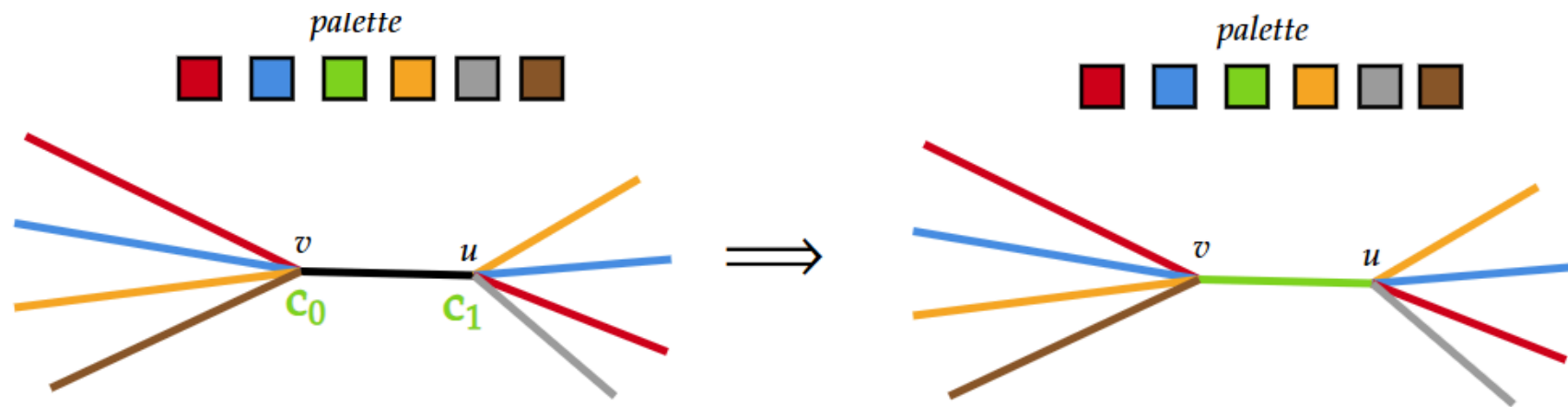
$$\chi'(G) \leq \Delta(G) + 1$$

proof:

- Proof by induction on $e(G)$.
- Pick $e = uv \in E(G)$ and set $G' := G - e$.
- By induction hypothesis, G' has a proper edge-colouring with at most $\Delta(G) + 1$ colours.
- Try to colour e when re-inserting it.

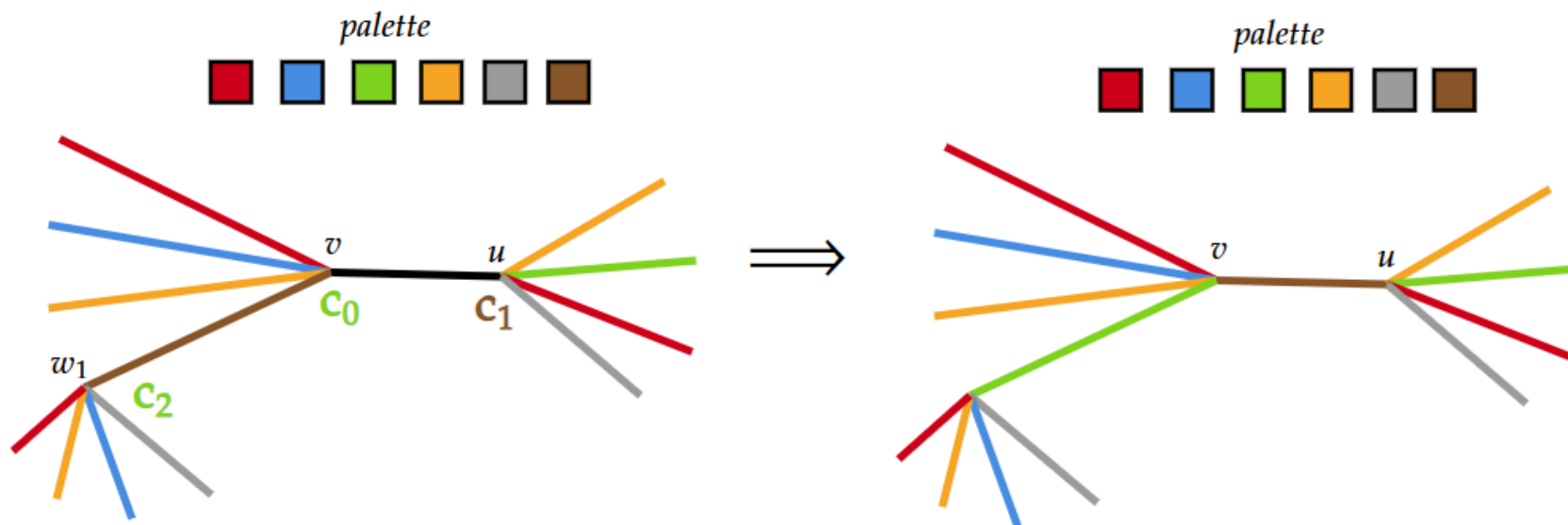


- Each endpoint misses at least one colour among $[\Delta(G) + 1]$.
- If there is a common missing colour at both u and v , assign it to $e := uv$.

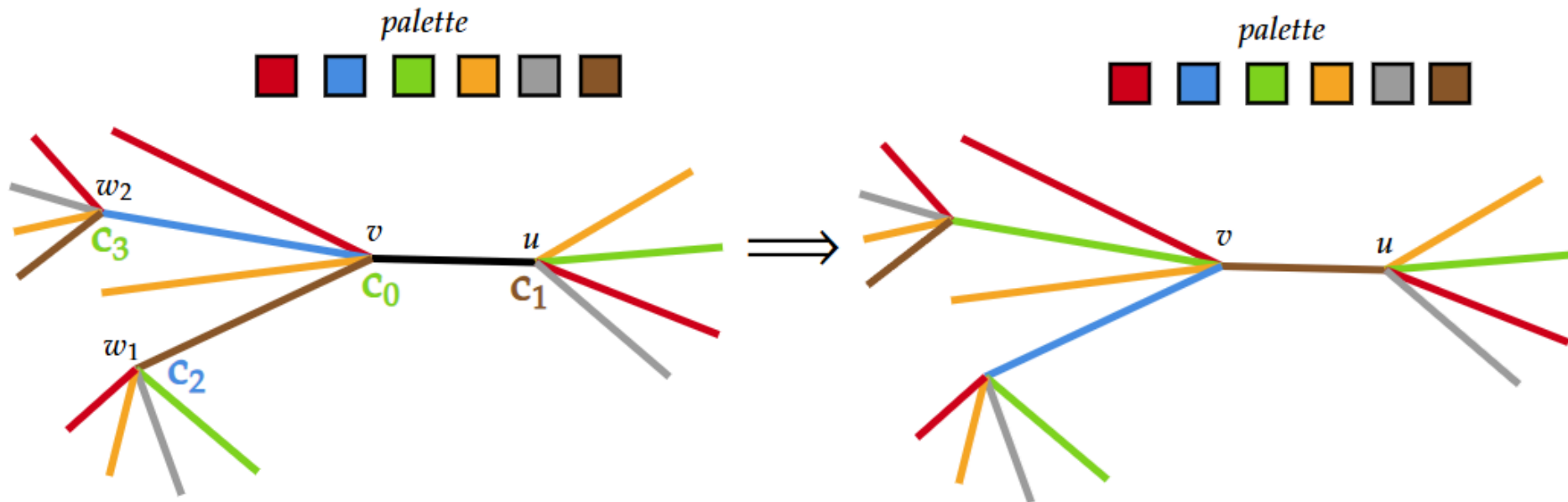


The palette is of size $\Delta(G) + 1$, which is 6 in this case

- Each endpoint misses at least one colour among $[\Delta(G) + 1]$.
- If there is no common missing colour at both u and v .
- Let c_1 be the colour that u don't see, and c_0 the colour that v don't see.
- Let vw_1 be the edge coloured in c_1 .
 - If w_1 also doesn't see c_0 colour vw_1 in c_0 , now v doesn't see c_1 so we can colour uv in c_1 .



- Let c_1 be the colour that u doesn't see, and c_0 the colour that v doesn't see.
- Let vw_1 be the edge coloured in c_1 .
- If w_1 also sees c_0 , let c_2 be the colour that w_1 doesn't see.
 - let vw_2 be an edge coloured in c_2 .
 - If w_2 doesn't see c_0 , then we can colour vw_2 in c_0 .
 - Now, v doesn't see the colour $\varphi(vw_2) = c_2$ so we can colour vw_1 in c_2
 - Finally, now v doesn't see the colour $\varphi(vw_1) = c_1$ so we can colour uv in c_1 .



We can do the process above again and again.

It remains to prove that this process stops, which is left for the **TA** session.

The intuition is that we always have 2 free colours per vertex so at some point those colours will start repeating.